Mehar Bawa *ML/AI Engineer*

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PROFILE

ML/AI Engineer experienced in designing and implementing **AI agent workflows**, **RAG** systems, evaluation frameworks and ML Models that transform raw data into actionable insights at Capgemini and CSU. Delivered production AI/ML solutions spanning classification, agentic workflows, RAG and forecasting systems.

EXPERIENCE

AI Engineer, *Colorado State University* 08/2026 – Present

- Architected & built a production **RAG** course AI assistant for 4 courses, 200 users on **Google Cloud** Run, spanning document ingestion, retrieval, LLM generation and a confidence-scoring system with role-based access control.
- Engineered a hybrid **FAISS + BM25** retrieval pipeline with reciprocal rank fusion and LLM reranking across per-course knowledge bases of up to 10k document chunks, achieving 91% mean confidence on in-scope questions and 100% correct refusal of off-syllabus queries.
- Implemented **Langfuse** for experiment tracking and prompt version control, enabling reproducibility and collaboration across prompt iterations and reducing debugging time by 20%.
- Built an LLM evaluation framework using prompt benchmarking, **LLM-as-judge** scoring and RAG metrics (Precision@K, Recall@K) with Responsible AI guardrails, improving response reliability by 18%.

Graduate Research Assistant, *Colorado State University* 02/2025 – 05/2026

- Engineered a dynamic-planning **LangGraph pipeline** using **OpenAI** and **Groq Llama** that auto-generates fine-grained classification taxonomies from unstructured data, classifying **109 cases** into 7 LLM-derived clusters.
- Designed a semantic filtering agent with embeddings-based vector retrieval, cutting inference costs by **~65%** through intelligent pre-filtering before downstream **RAG** processing.
- Architected a **ReAct**-based RAG pipeline using **GPT-5**, interleaving reasoning and tool-calling steps with **vector-based retrieval** across enterprise knowledge repositories to deliver structured outputs with **100% citation accuracy** for research applications.

Machine Learning Engineer, *Capgemini* 03/2023 – 01/2025 | Noida, IN

- Forecasted demand for ~1800 SKUs by deploying a **4-tier** routed **time-series** system using **CatBoost** quantile regression, routing each SKU by sales history at both SKU and stylecode level.
- Built a multi-channel **data pipeline** ingesting orders from **3 sales channels** across 8 country-specific data collections, applying stockout-adjusted demand correction capped at **2x** inflation to estimate true demand.
- Solved the **cold-start** problem for **~40%** of products by designing a **launch curve** that predicts a new SKU's sales trajectory before it has any sales history, blending it with real data as the SKU matures.

Data & Cloud Consultant, *Capgemini* 02/2020 – 02/2023

- Eliminated **20+ hours** of monthly manual intervention by building an **AWS Lambda** event-driven automation pipeline for platform monitoring and maintenance, sustaining **100% uptime** through automated remediation.
- Decreased system failures by **70%** by applying FMEA methodology across cross-regional teams in a fast-paced, pilot-driven environment, surfacing insights in **Power BI** dashboards to drive root-cause remediations.
- Reduced deployment lead time by **40%** by implementing **CI/CD pipelines** and **Terraform IaC** on AWS ECS, increasing reusability of code and design patterns through automated versioning, rollback, and testing.

MACHINE LEARNING & AI PROJECTS

Agentic AI Resume Screening System

- Developed and iterated on an **agentic LLM system** using LangChain and **LangGraph** with structured tool use and chained workflows, improving matching accuracy by **~30%**.
- Implemented agentic AI-driven quality assurance through a **reflection loop** with a Critic agent and **Pydantic** guardrails, reducing LLM hallucination rate by **~25%.**
- Established monitoring and observability using **LangSmith**, keeping production LLM workflows healthy by tracking latency, cost, and failure modes.

Financial Sentiment Analysis using FinBERT

- Prototyped with fine-tuning techniques on **FinBERT** using **PyTorch** and **HuggingFace**, achieving **96.5% F1** and 100% precision through rigorous evaluation and iteration.

Diabetes Prediction Project

- Achieved **82.1% ROC-AUC** by developing a clinical prediction model using **LightGBM** and **XGBoost**, with rigorous feature engineering and evaluation on real-world healthcare data.
- Built a production-ready Python ML application using **FastAPI** and **MLflow**, supporting model testing, deployment, and monitoring with automated CI/CD.
- Orchestrated an AI dietary-guidance system by integrating **CrewAI** with a custom **Model Context Protocol (MCP)** server, enabling discoverable tool access between an LLM agent and a scikit-learn classifier.

SKILLS & CERTIFICATIONS

ML / AI: Python (pandas, scikit-learn), LangChain, LangGraph, OpenAI API, RAG, NLP, LLM, PyTorch/TensorFlow, HuggingFace Transformers, Claude Code, MCP, Vector Databases, CrewAI, PySpark, Cursor
MLOps, Deployment & BI: FastAPI, Streamlit, MLflow, Docker, CI/CD, Git/GitHub, Power BI/Tableau
Cloud & Data: AWS (Lambda, ECS, EC2, ELB, S3, EBS, EKS, CloudWatch, IAM), Snowflake, Terraform, SQL, PostgreSQL, GCP, Databricks
Certifications: AWS Certified Developer - Associate, Certified SAFe 6 Product Owner

EDUCATION

MS in Computer Information Systems, *Colorado State University* 01/2025 – 05/2026 | GPA: 3.94

- **Teaching Assistant** for Analytics and AI in Business & Member of **Dean's List** for Spring 2025, Fall 2025.

B-Tech in Computer Science and Engineering, *SRM University* 05/2019